

MBAi 448 | Winter 2026

AI Today & Beyond

Attendance

Enter the “Magic Word” in Canvas

Do Not Share with Absent Students

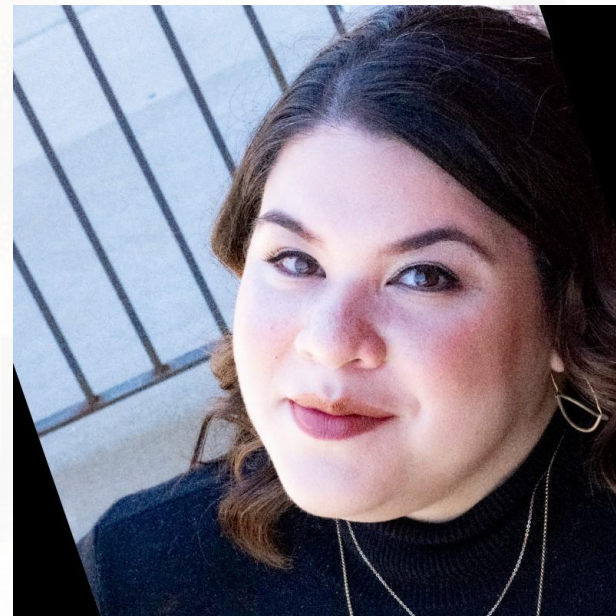


Our Judges



Armando Pauker

Co-Founder & Managing Director
Tensility Venture Partners



Ellie Bahrmasel

Founder
Stealth Mode



Erik Brown

Senior Director, Private Equity
Northwestern Kellogg



Oleg Evdokimov

Group Manager, Product Design
Relativity



The AI of Today



Everything Covered This Quarter





The AI of Tomorrow

Frontier Challenges and Potential Future Capabilities

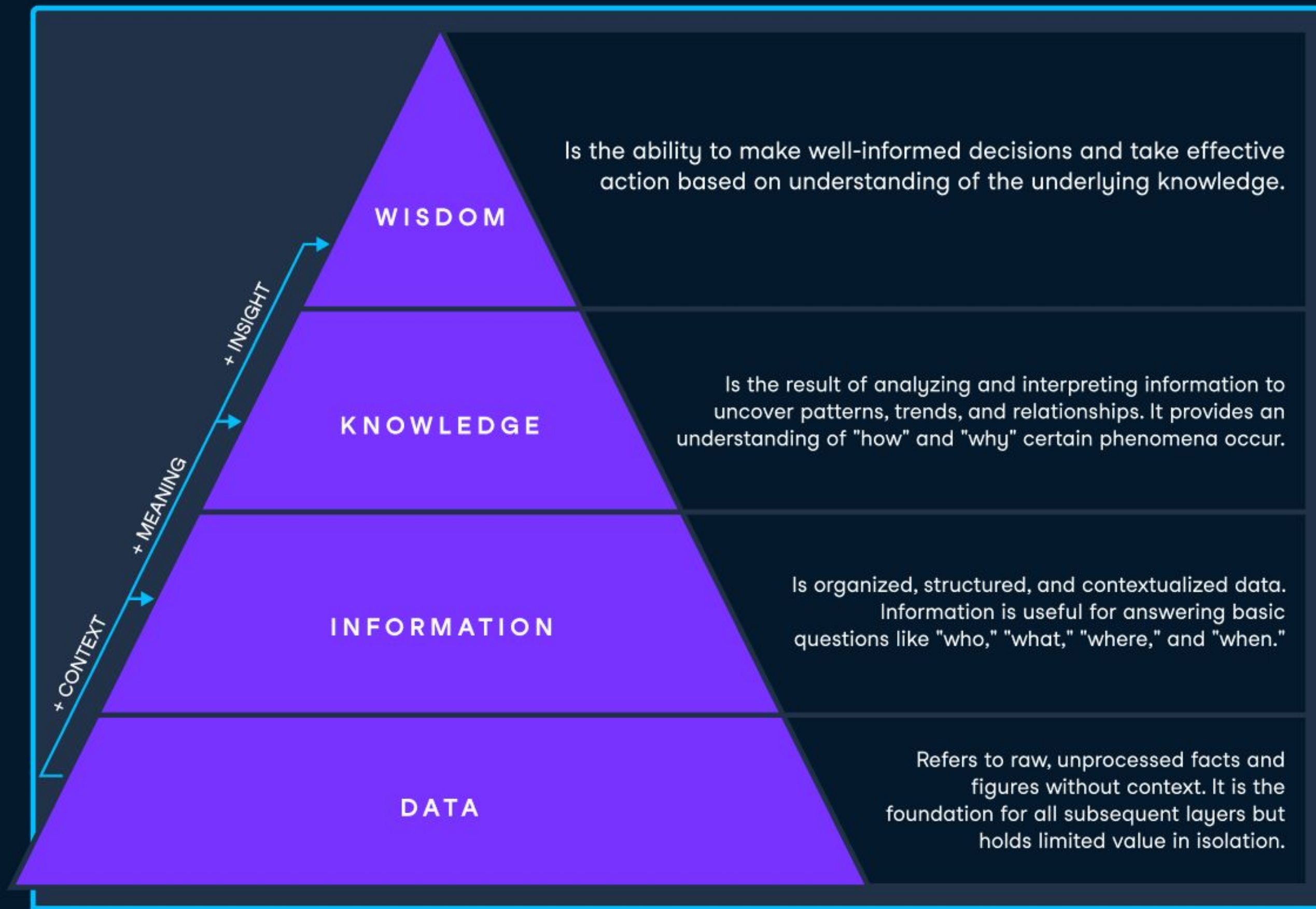
What if AI could “perceive” and
“understand” multiple modalities at
once? Example Use Cases?



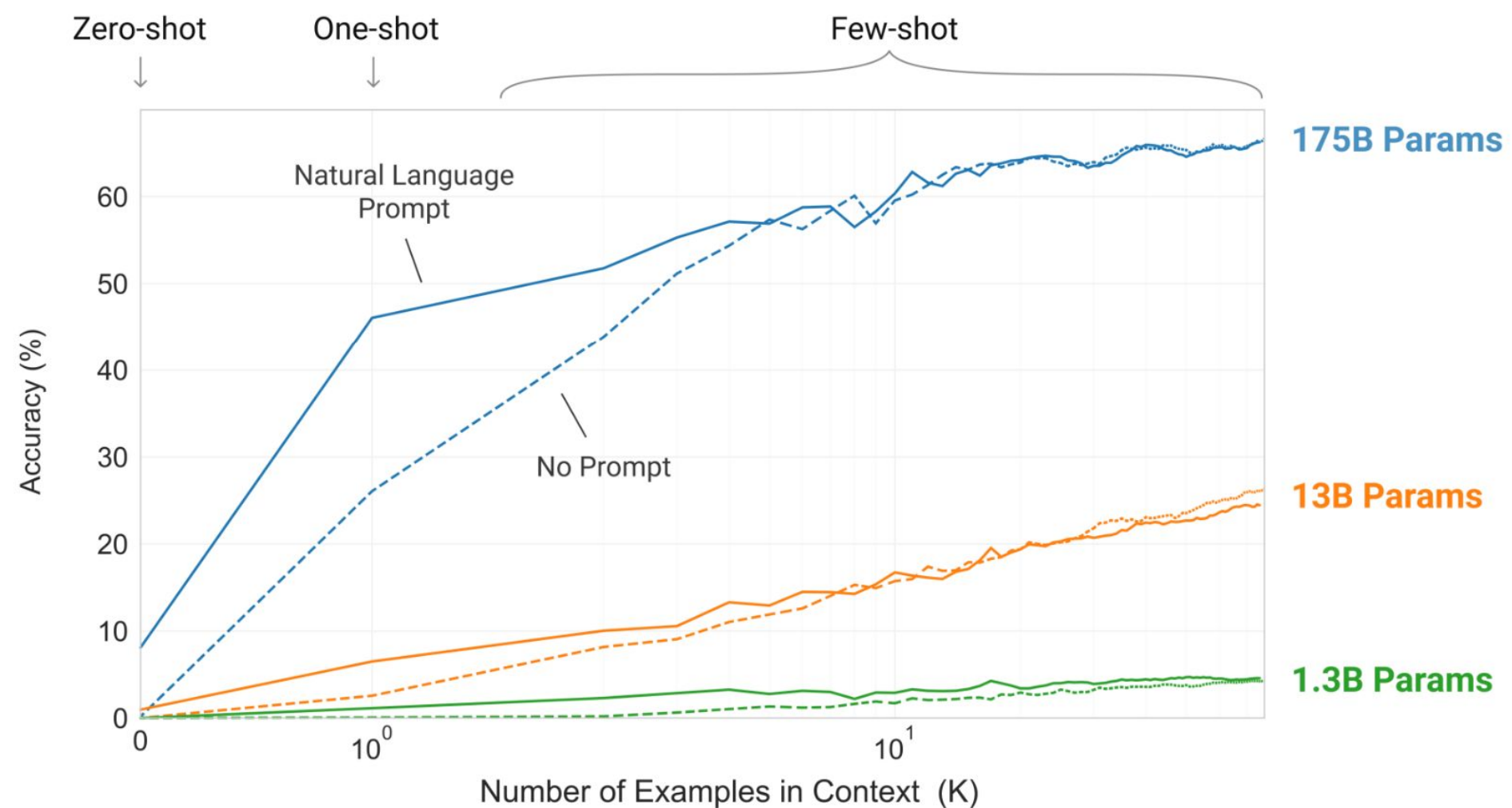
What if AI could “understand” and generate multiple modalities at once? Example Use Cases?

Your tools
should help
you work
better.



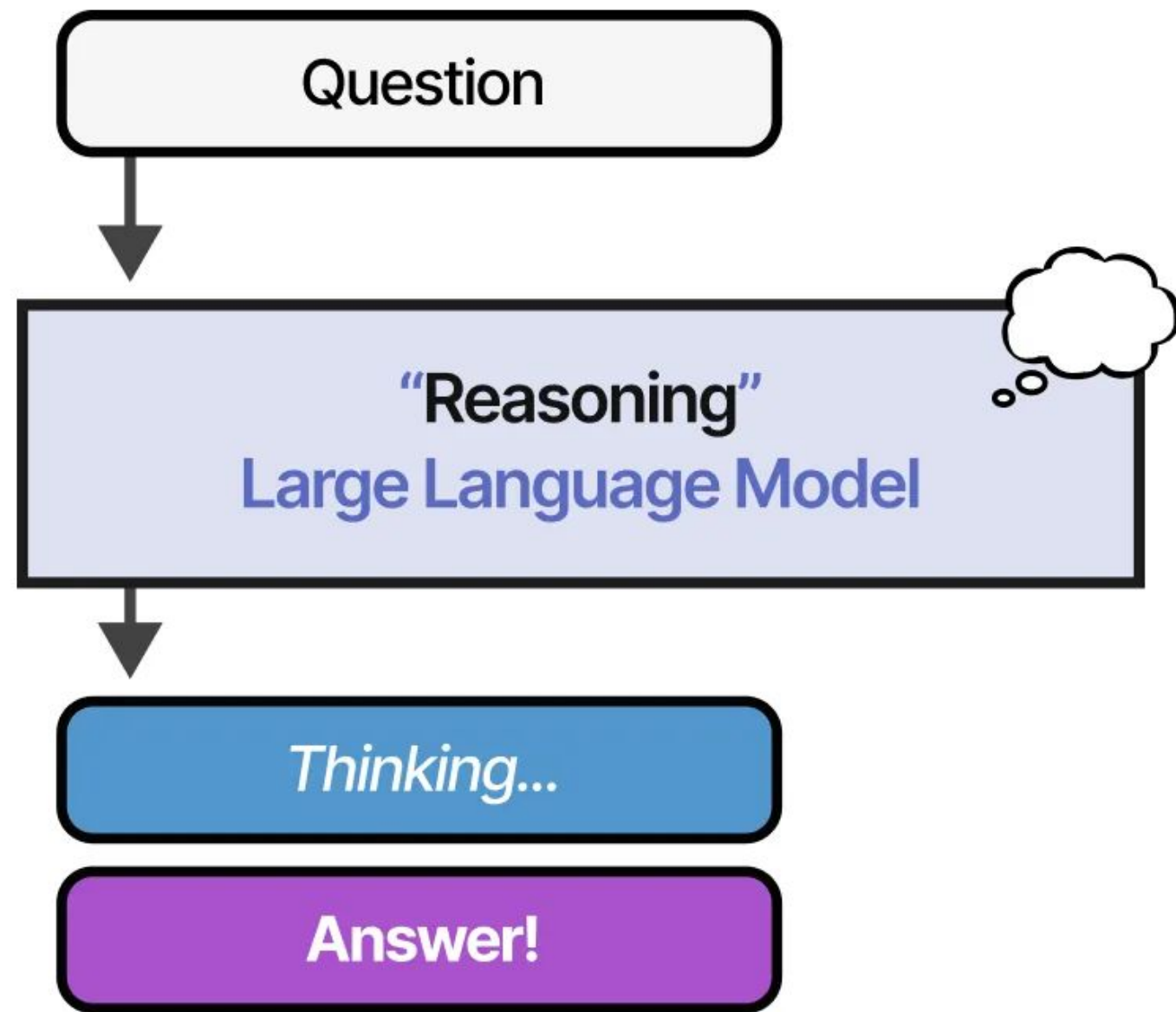


Scaling Deep Learning + Symbolic AI



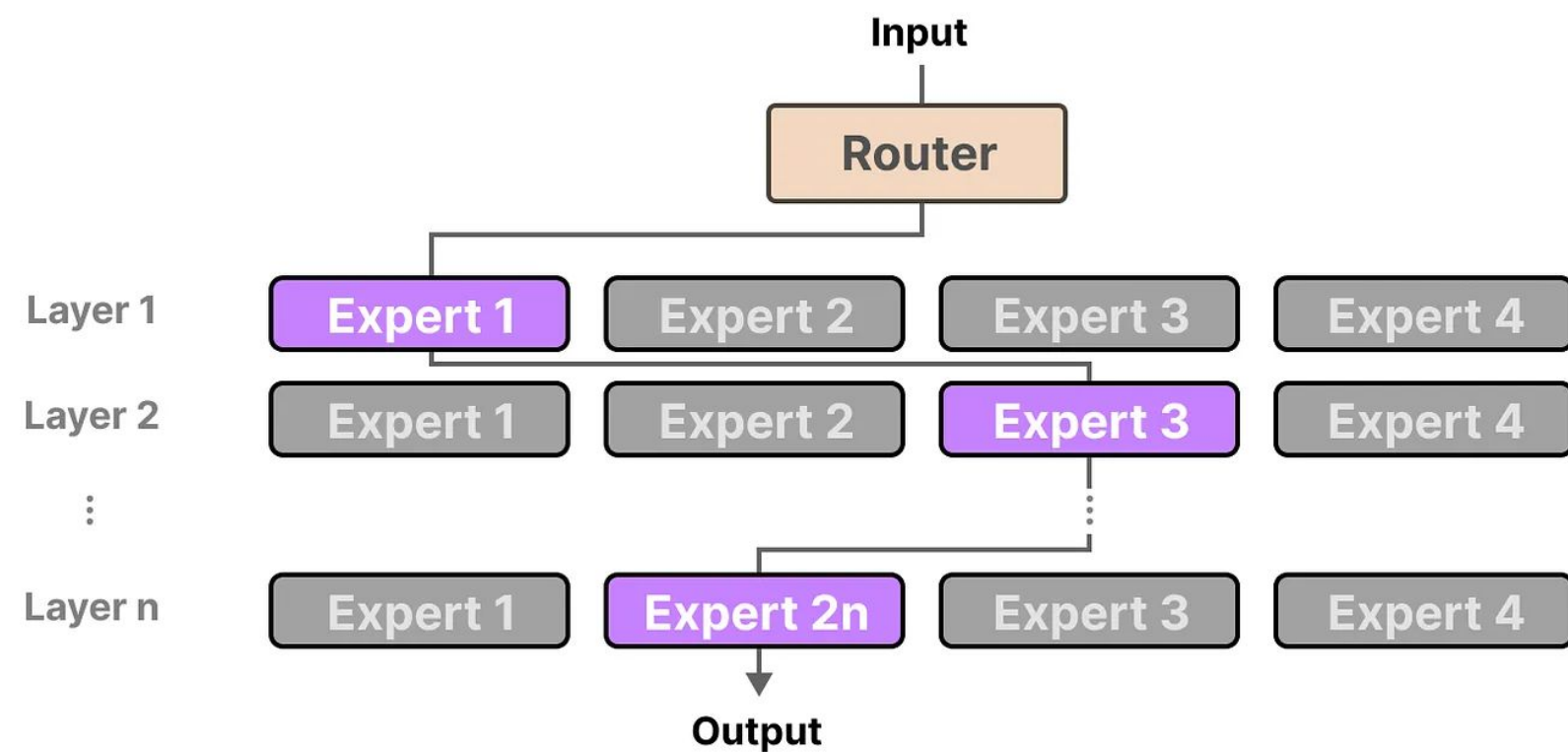
Combining neural networks with logic-based reasoning to build AI systems that can handle both pattern recognition and explicit rules.

Chain of Thought & Reasoning



Prompting techniques that guide AI to show its reasoning process step-by-step, improving problem-solving accuracy and transparency.

Mixture of Experts



Mixture of Experts boosts AI performance by activating specialized models for specific tasks, improving efficiency and scalability.

Knowledge Distillation & Small Models

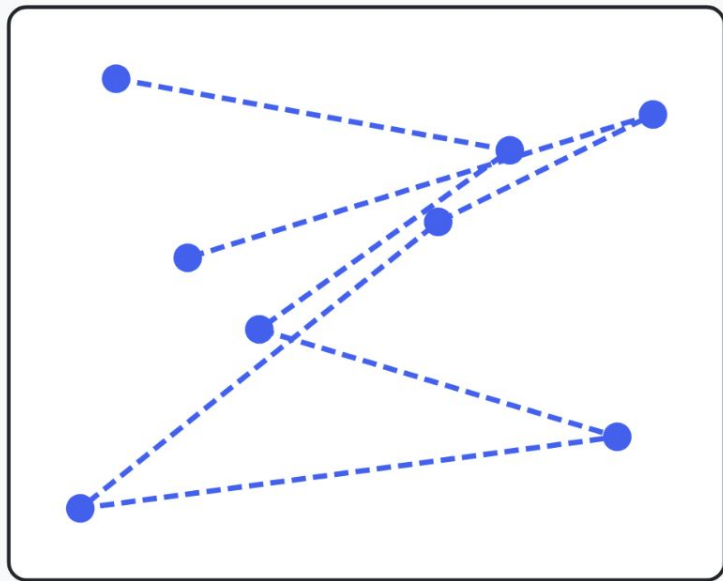


Transferring knowledge from large AI models to smaller ones, creating efficient, faster systems that maintain quality while reducing computing needs.

Search, Sampling, and Entropy

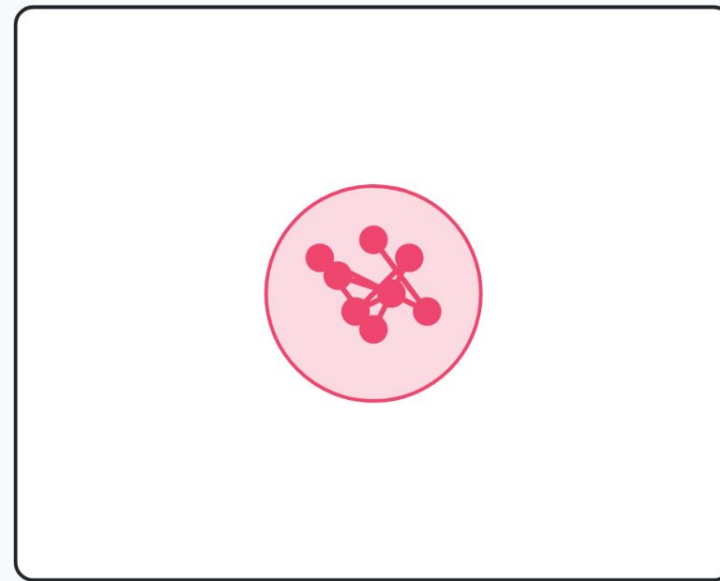
Exploration vs. Exploitation

EXPLORATION



- Discovering new areas
- High uncertainty
- Long-term benefits

EXPLOITATION

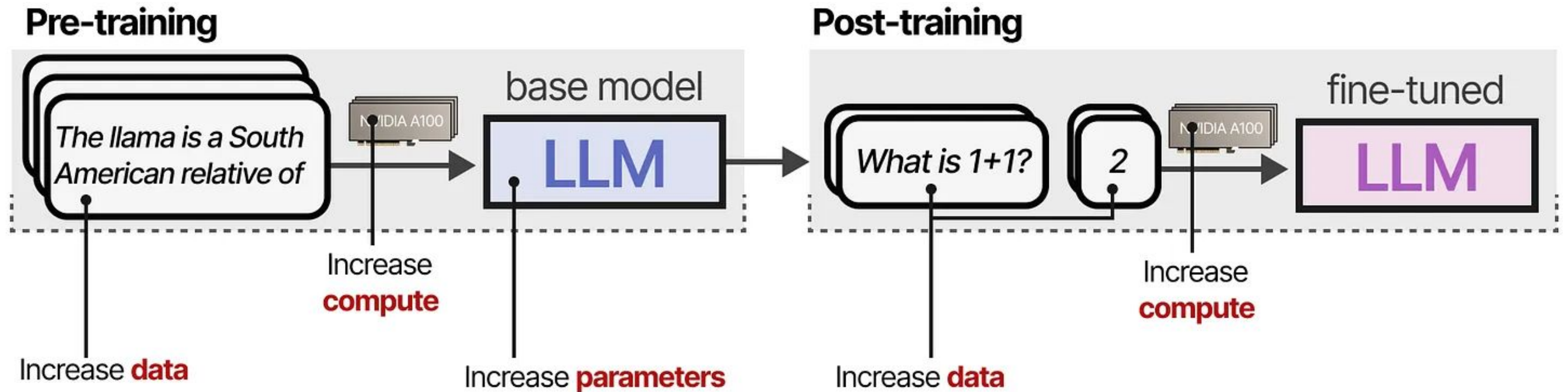


- Refining known solutions
- Low uncertainty
- Immediate rewards

Algorithms that efficiently explore solution spaces, critical for helping AI systems gain general intelligence capabilities (e.g., Monte Carlo Tree Search and Beam Search).

What's the difference between learning and search?

Training-time Compute



Test-time Compute

“Reasoning” LLMs

Pre-training

Fine-tuning

Inference

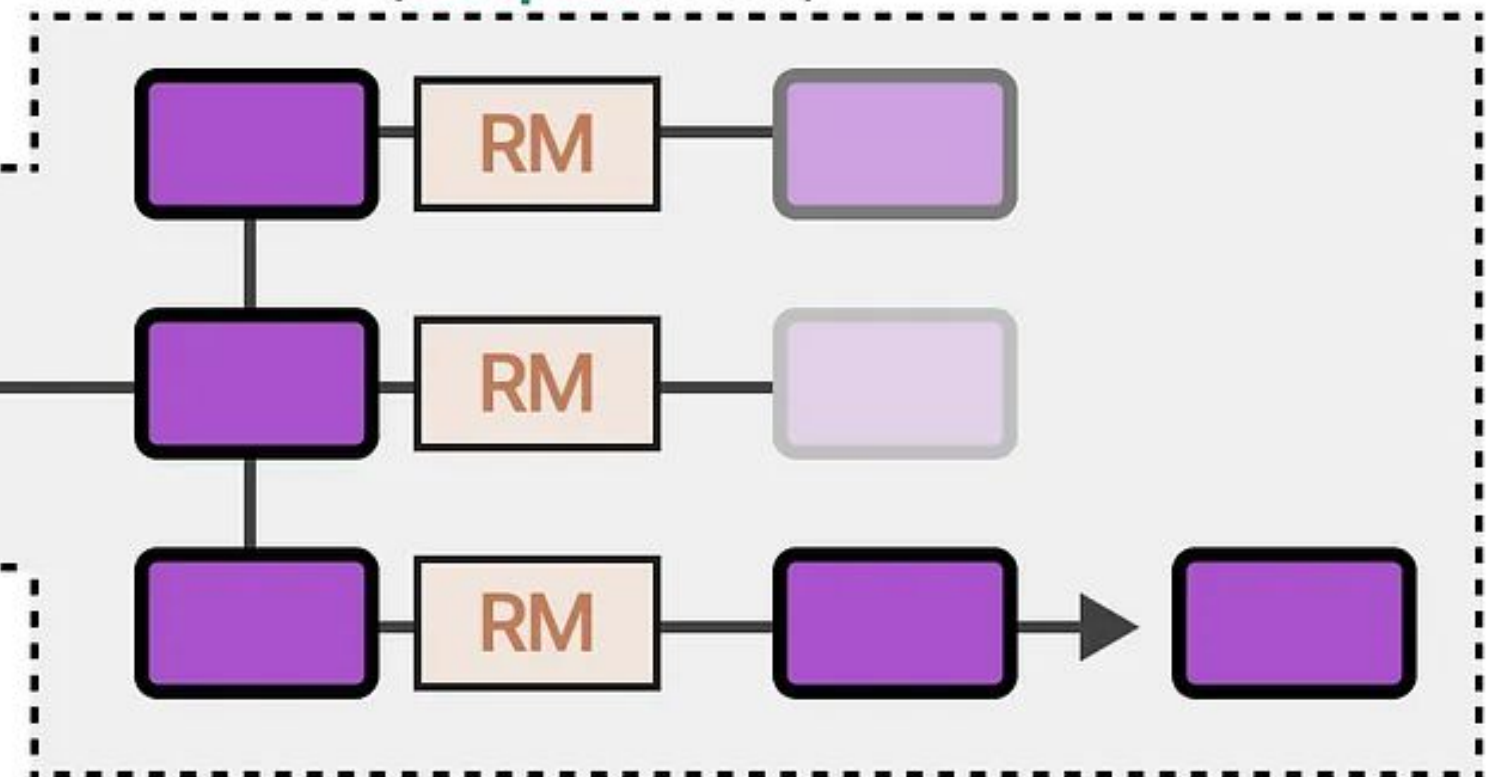
Test-time compute

training (input level)



Modifying Proposal Distribution
(learned behavior)

inference (output level)



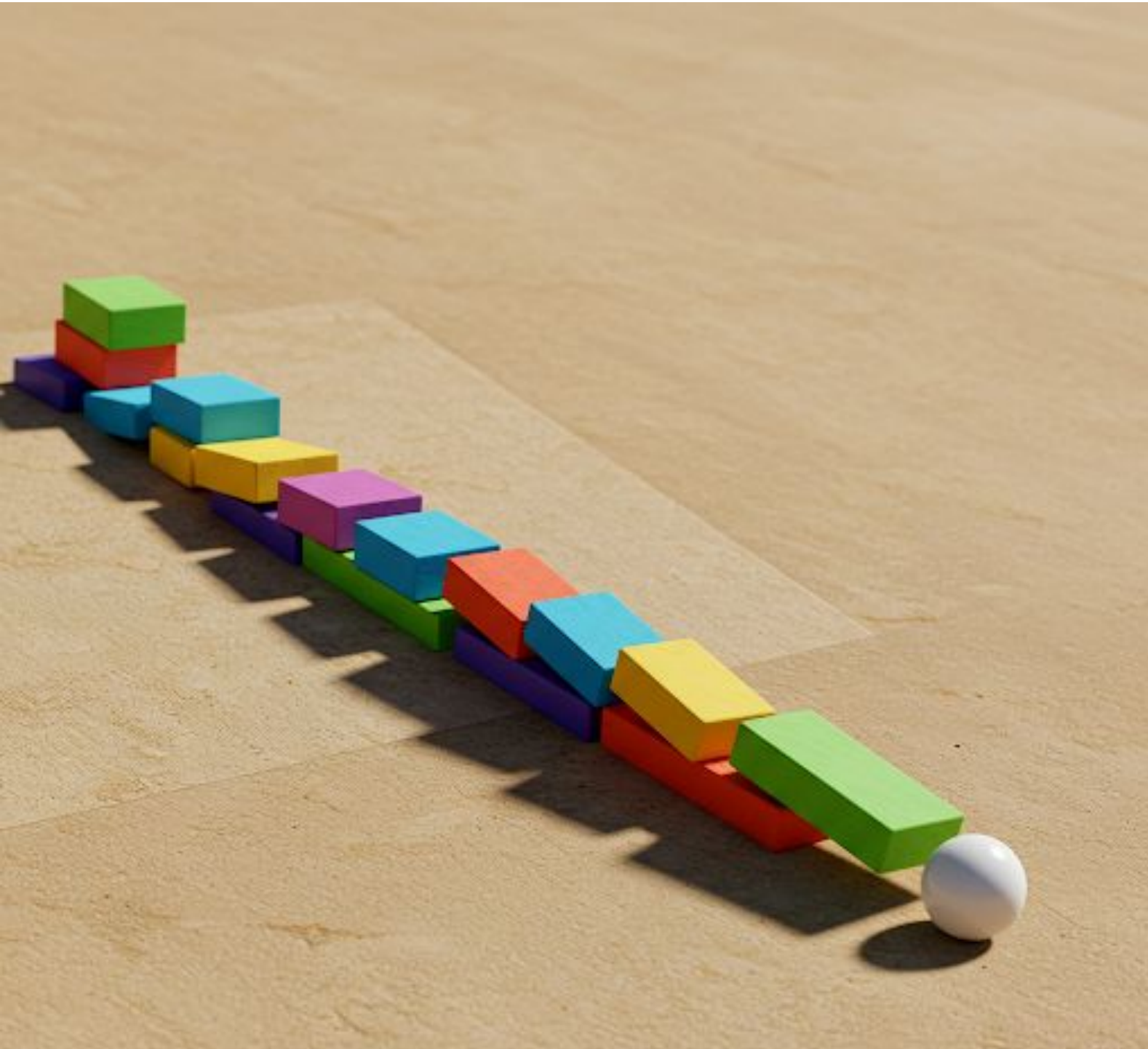
Search against Verifiers
(found behavior)

Curiosity, Inventive Creativity, & Novelty



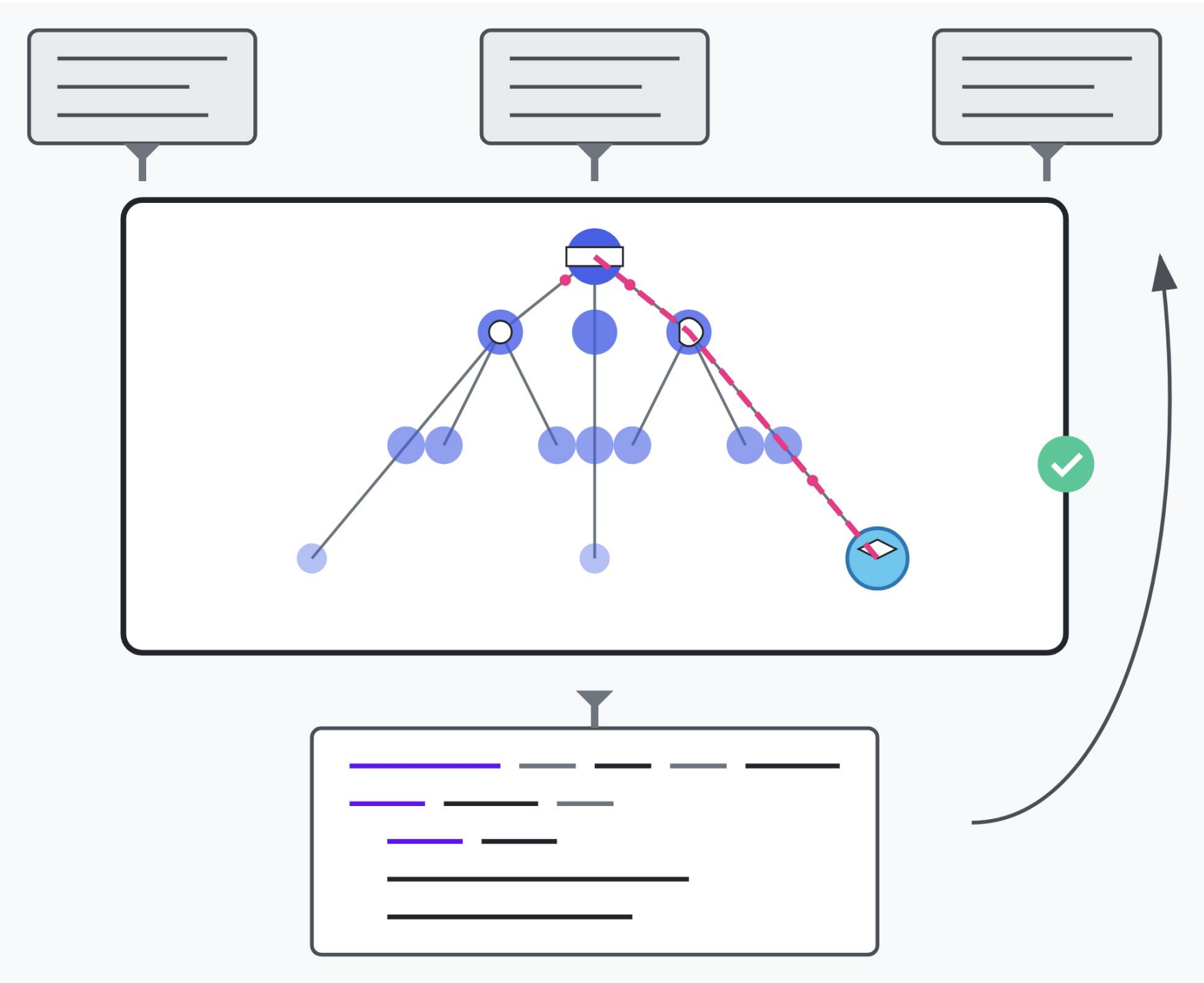
Driving AI to explore new ideas, create unique solutions, and push beyond existing knowledge and capabilities.

Causality, Planning, & Agency



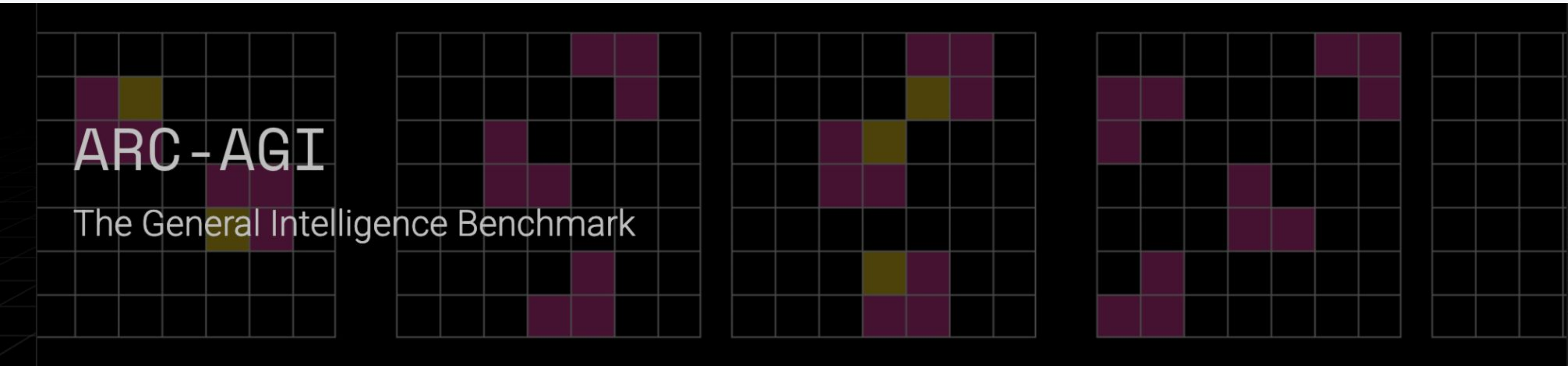
Helping AI understand cause-effect relationships, make strategic decisions, and act with goal-directed behavior.

Abstractions & Synthesis



Enabling AI to simplify complex information and combine ideas, mimicking how humans understand and solve problems.

AGI Definition per ARC-AGI



AGI is a system that can efficiently acquire new skills outside of its training data.

Easy for Humans, Hard for AI



ARC PRIZE REMAINS UNDEFEATED.
NEW IDEAS STILL NEEDED

- > Home
- > ARC-AGI
- > 2024 Results
- > Technical Guide
- > Play
- > Blog

2025 SIGN UP

OpenAI o3 unlocks breakthrough high score on ARC-AGI-Pub. [Learn more.](#)

ARC PRIZE

ARC Prize is a \$1,000,000+ public competition to beat and open source a solution to the ARC-AGI benchmark.

Hosted by [Mike Knoop](#) (Co-founder, Zapier) and [François Chollet](#) (Creator of ARC-AGI, Keras).

ARC Prize 2025 coming soon.

> [See 2024 winners](#)

ARC-AGI

Most AI benchmarks measure skill. But skill is not intelligence. General intelligence is the ability to efficiently acquire new skills. Chollet's unbeaten 2019 Abstraction and Reasoning Corpus for Artificial General Intelligence ([ARC-AGI](#)) is the only formal benchmark of AGI progress.

It's easy for humans, but hard for AI.

PLAY

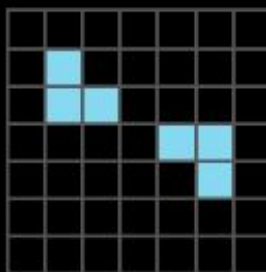
Try ARC-AGI. Given the examples, identify the pattern and solve the test puzzle.

Puzzle ID: 3aa6fb7a

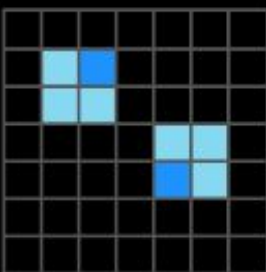
Previous 1 of 6 Next

EXAMPLES

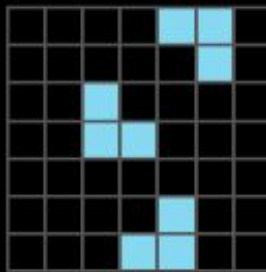
Ex.1 Input (7x7)



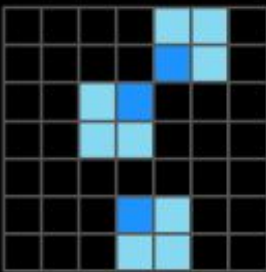
Ex.1 Output (7x7)



Ex.2 Input (7x7)

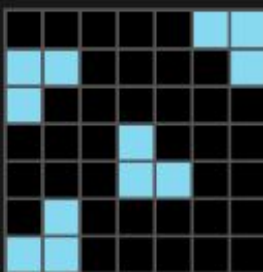


Ex.2 Output (7x7)

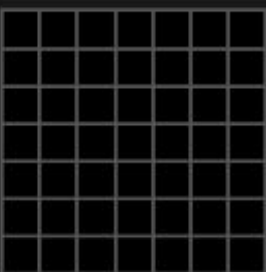


TEST

Input (7x7)



Output (7x7)



1. Configure your output grid:

Copy from input

Reset

2. Click to select a color:



3. See if your output is correct:

Submit solution

Demo

Feel free to follow along!

Artificial General Intelligence (AGI)

Goals:

- Explore the ARC-AGI Benchmark and ARC-AGI Prize to learn more about the current progress towards AGI

Links:

- [ARC Prize](#)

Reflect + Discuss:

- Share your thoughts

Just A Rather Very Intelligent System



Researching Emerging & State-of-the-Art AI



Artificial Intelligence

Authors and titles for recent submissions

- [Wed, 21 Aug 2024](#)
- [Tue, 20 Aug 2024](#)
- [Mon, 19 Aug 2024](#)
- [Fri, 16 Aug 2024](#)
- [Thu, 15 Aug 2024](#)

See today's [new](#) changes

Total of 611 entries : 1–25 [26–50](#) [51–75](#) [76–100](#) ... [601–611](#)

Showing up to 25 entries per page: [fewer](#) | [more](#) | [all](#)

Wed, 21 Aug 2024 (showing first 25 of 164 entries)

[1] [arXiv:2408.11042](#) [[pdf](#), [html](#), [other](#)]

GraphFSA: A Finite State Automaton Framework for Algorithmic Learning on Graphs

[Florian Grötschla](#), [Joël Mathys](#), [Christoffer Raun](#), [Roger Wattenhofer](#)

Comments: Published as a conference paper at ECAI 2024

Subjects: [Artificial Intelligence](#) (cs.AI)

[2] [arXiv:2408.11039](#) [[pdf](#), [html](#), [other](#)]

Transfusion: Predict the Next Token and Diffuse Images with One Multi-Modal Model

[Chunting Zhou](#), [Lili Yu](#), [Arun Babu](#), [Kushal Tirumala](#), [Michihiro Yasunaga](#), [Leonid Shamis](#), [Jacob Kahn](#), [Xuezhe Ma](#), [Luke Zettlemoyer](#), [Omer Levy](#)

Comments: 23 pages

Subjects: [Artificial Intelligence](#) (cs.AI); [Computer Vision and Pattern Recognition](#) (cs.CV)

[3] [arXiv:2408.10970](#) [[pdf](#), [html](#), [other](#)]

Hybrid Recurrent Models Support Emergent Descriptions for Hierarchical Planning and Control

[Poppy Collis](#), [Ryan Singh](#), [Paul F Kinghorn](#), [Christopher L Buckley](#)

Comments: 4 pages, 3 figures

Subjects: [Artificial Intelligence](#) (cs.AI); [Systems and Control](#) (eess.SY)

[4] [arXiv:2408.10947](#) [[pdf](#), [html](#), [other](#)]

Dr.Academy: A Benchmark for Evaluating Questioning Capability in Education for Large Language Models

[Yuyan Chen](#), [Chenwei Wu](#), [Songzhou Yan](#), [Panjun Liu](#), [Haoyu Zhou](#), [Yanghua Xiao](#)

The Thirty-eighth Annual Conference on Neural Information Processing Systems

NeurIPS 2024

 Vancouver, Canada  Dec 16 2024  <https://neurips.cc/>  pc2024@neurips.cc

For Authors
Please see our [call for papers](#) with detailed instructions before you submit.
You must format your submission using the [NeurIPS 2024 LaTeX style file](#) or the [NeurIPS 2024 template on Overleaf](#).

IMPORTANT:
Please join the NeurIPS 2024 Checklist Assistant Study that will provide you with free verification of your checklist performed by an LLM: [form link](#), see [our blog](#) for details.
Submission Start: Apr 22 2024 08:00PM UTC-0, Abstract Registration: May 15 2024 08:00PM UTC-0, Submission Deadline: May 22 2024 08:00PM UTC-0

Accept (oral)	Accept (spotlight)	Accept (poster)	Reject	Recent Activity
Learning to grok: Emergence of in-context learning and skill composition in modular arithmetic tasks  Tianyu He, Darshil Doshi, Aritra Das, Andrey Gromov Published: 25 Sept 2024, Last Modified: 06 Nov 2024 NeurIPS 2024 oral Readers:  Everyone 14 Replies Show details Trading Place for Space: Increasing Location Resolution Reduces Contextual Capacity in Hippocampal Codes  Spencer Rooke, Zhaoze Wang, Ronald W Di Tullio, Vijay Balasubramanian Published: 25 Sept 2024, Last Modified: 10 Jan 2025 NeurIPS 2024 oral Readers:  Everyone 11 Replies Show details Visual Autoregressive Modeling: Scalable Image Generation via Next-Scale Prediction  Keyu Tian, Yi Jiang, Zehuan Yuan, BINGYUE PENG, Liwei Wang Published: 25 Sept 2024, Last Modified: 16 Jan 2025 NeurIPS 2024 oral Readers:  Everyone 30 Replies Show details Cracking the Code of Juxtaposition: Can AI Models Understand the Humorous Contradictions  Zhe Hu, Tuo Liang, Jing Li, Yiren Lu, Yunlai Zhou, Yiran Qiao, Jing Ma, Yu Yin Published: 25 Sept 2024, Last Modified: 06 Nov 2024 NeurIPS 2024 oral Readers:  Everyone 26 Replies Show details Human Expertise in Algorithmic Prediction  Rohan Alur, Manish Raghavan, Devavrat Shah Published: 25 Sept 2024, Last Modified: 06 Nov 2024 NeurIPS 2024 oral Readers:  Everyone 15 Replies Show details Learning diffusion at lightspeed  Antonio Terpin, Nicolas Lanzetti, Martín Gadea, Florian Dorfler Published: 25 Sept 2024, Last Modified: 06 Nov 2024 NeurIPS 2024 oral Readers:  Everyone 13 Replies Show details Weisfeiler and Leman Go Loopy: A New Hierarchy for Graph Representational Learning  Raffaele Paolino, Sohir Maskey, Pascal Welke, Gitta Kutyniok Published: 25 Sept 2024, Last Modified: 06 Nov 2024 NeurIPS 2024 oral Readers:  Everyone 12 Replies Show details Unlocking the Capabilities of Thought: A Reasoning Boundary Framework to Quantify and Optimize Chain-of-Thought  Qiguang Chen, Libo Qin, Jiaqi WANG, Jingxuan Zhou, Wanxiang Che Published: 25 Sept 2024, Last Modified: 25 Dec 2024 NeurIPS 2024 oral Readers:  Everyone 23 Replies Show details				

NEW Welcome Fireworks.ai on the Hub



The AI community building the future.

The platform where the machine learning community collaborates on models, datasets, and applications.

[Explore AI Apps](#)
[or Browse 1M+ models](#)

Tasks Libraries Datasets Languages Licenses Other

Multimodal

- Text-to-Image
- Image-to-Text
- Text-to-Video
- Visual Question Answering
- Document Question Answering
- Graph Machine Learning

Computer Vision

- Depth Estimation
- Image Classification
- Object Detection
- Image Segmentation
- Image-to-Image
- Unconditional Image Generation
- Video Classification
- Zero-Shot Image Classification

Natural Language Processing

- Text Classification
- Token Classification
- Table Question Answering
- Question Answering
- Zero-Shot Classification
- Translation
- Summarization
- Conversational
- Text Generation
- Text2Text Generation
- Sentence Similarity

Audio

- Text-to-Speech
- Automatic Speech Recognition
- Audio-to-Audio
- Audio Classification
- Voice Activity Detection

Tabular

- Tabular Classification
- Tabular Regression

Reinforcement Learning

- Reinforcement Learning
- Robotics

Models 469,541

meta-llama/Llama-2-70b
Text Generation • Updated 4 days ago • 25.2k • 64

stabilityai/stable-diffusion-xl-base-0.9
Updated 6 days ago • 2.01k • 393

openchat/openchat
Text Generation • Updated 2 days ago • 1.3k • 136

llyasviel/ControlNet-v1-1
Updated Apr 26 • 1.87k

cerspense/zeroscope_v2_XL
Updated 3 days ago • 2.66k • 334

meta-llama/Llama-2-13b
Text Generation • Updated 4 days ago • 328 • 64

tiiuae/falcon-40b-instruct
Text Generation • Updated 27 days ago • 288k • 899

WizardLM/WizardCoder-15B-V1.0
Text Generation • Updated 3 days ago • 12.5k • 332

CompVis/stable-diffusion-v1-4
Text-to-Image • Updated about 17 hours ago • 448k • 5.72k

stabilityai/stable-diffusion-2-1
Text-to-Image • Updated about 17 hours ago • 782k • 2.81k

Salesforce/xgen-7b-8k-inst
Text Generation • Updated 4 days ago • 6.18k • 57

<https://huggingface.co>

kaggle

+

Create

Home

Competitions

Datasets

Models

Code

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ABSTRACTION AND REASONING CORPUS · FEATURED CODE COMPETITION · 3 MONTHS AGO

Late Submission



ARC Prize 2024

Create an AI capable of solving reasoning tasks it has never seen before

Overview

Data

Code

Models

Discussion

Leaderboard

Rules

Overview

In this competition, you'll develop AI systems to efficiently learn new skills and solve open-ended problems, rather than depend exclusively on AI systems trained with extensive datasets. The top submissions will show improvement toward human reasoning benchmarks.

Start

Jun 11, 2024

Close

Nov 10, 2024

Merger & Entry

Competition Host

Abstraction and Reasoning Corpus



Prizes & Awards

\$1,100,000

Awards Points & Medals

Participation

16,349 Entrants

1,637 Participants

1,427 Teams

16,956 Submissions

https://www.kaggle.com

Keepin' Up With AI



OpenAI

ANTHROPIC

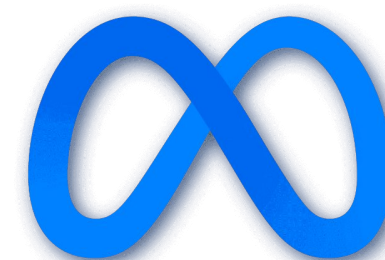


DeepMind

Microsoft®
Research



Google AI

The Meta logo, featuring a blue infinity symbol.
Meta



Hugging Face

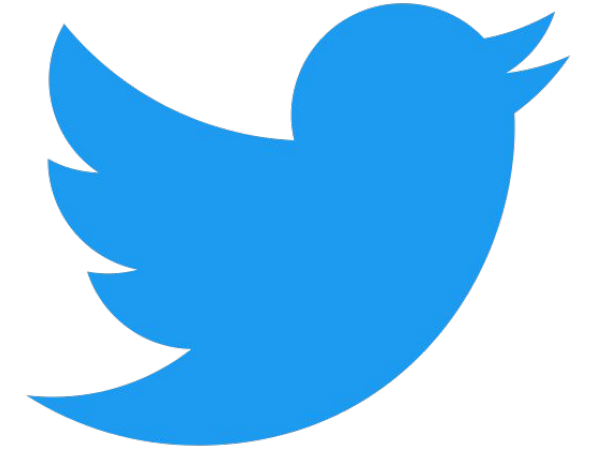
Radar

Radar Trends

A monthly list of technology and business trends we find important or interesting.



Courses



Google
Alerts



Papers With Code

Blogs

LinkedIn

arXiv



Why of AI[®]

The image is a large grid of logos for various AI and data companies, organized into several categories. The categories are as follows:

- INFRASTRUCTURE:** Includes logos for storage (AWS, IBM, VAST, etc.), data lakes/warehouses (Dremio, Snowflake, etc.), streaming/in-memory (AWS, etc.), data warehouses (AWS, etc.), real-time databases (CockroachDB, etc.), graph databases (Neo4j, etc.), GPU databases (Kinetica, etc.), multi-model databases (Couchbase, etc.), and ETL/ELT (dbt, etc.).
- ANALYTICS:** Includes BI platforms (Looker, etc.), visualization (Tableau, etc.), data science notebooks (Jupyter, etc.), data science platforms (IBM, etc.), enterprise ML/AI platforms (Databricks, etc.), data generation & labeling (Scale, etc.), and data analyst platforms (Mode, etc.).
- MACHINE LEARNING & ARTIFICIAL INTELLIGENCE:** Includes MLOps (Weights & Biases, etc.), AI observability (Fiddler, etc.), AI safety & security (Giskard, etc.), commercial AI research (OpenAI, etc.), non-profit AI research (EleutherAI, etc.), AI hardware (Intel, etc.), GPU cloud/infra (Databricks, etc.), edge AI (Hailo, etc.), closed source models (OpenAI, etc.), and AI frameworks/tools/libraries (PyTorch, etc.).
- APPLICATIONS - ENTERPRISE:** Includes sales (HubSpot, etc.), marketing (Klaviyo, etc.), customer experience (Intercom, etc.), human capital (Gigamonks, etc.), automation & operations (UiPath, etc.), decision & optimization (Nextmv, etc.), legal (eScrip, etc.), partnerships (Partnership, etc.), retech & compliance (Data Vanta, etc.), finance (Anaplan, etc.), and finance & insurance (Kensho, etc.).
- APPLICATIONS - HORIZONTAL:** Includes code & documentation (Explit, etc.), text (Jasper, etc.), audio & voice (Eleven Labs, etc.), image (Midjourney, etc.), presentation & design (Figma, etc.), video editing (Descript, etc.), animation & 3D/gaming (Unreal, etc.), and search/conversional AI (Search, etc.).
- APPLICATIONS - INDUSTRY:** Includes healthcare (Tempus, etc.), life sciences (Moderna, etc.), transportation (Uber, etc.), agriculture (John Deere, etc.), industrial & logistics (Bosch, etc.), aerospace/defense & govt (Andur, etc.), and cross-industry (Palantir, etc.).
- OPEN SOURCE INFRASTRUCTURE:** Includes data frameworks (Spark, etc.), formats (JSON, etc.), query/data flow (Airflow, etc.), data management (Amundsen, etc.), databases (PostgreSQL, etc.), OLAP (DuckDB, etc.), orchestration (Apache Airflow, etc.), infra-structure (Kubernetes, etc.), streaming & messaging (Kafka, etc.), stat tools & languages (Python, etc.), MLOps & AI infra (Weights & Biases, etc.), AI frameworks/tools/libraries (PyTorch, etc.), and AI models (OpenAI, etc.).
- DATA SOURCES & APIs:** Includes data marketplaces & discovery (AWS, etc.), financial & market data (Bloomberg, etc.), air/space/sea (Orbital Insight, etc.), people/entities (Zoom, etc.), location intelligence (FourSquare, etc.), ESG (Sustainalytics, etc.), data & AI consulting (QuantumBlack, etc.), and data sources & APIs (Bloomberg, etc.).

THE TOP 100

GEN AI >>>>

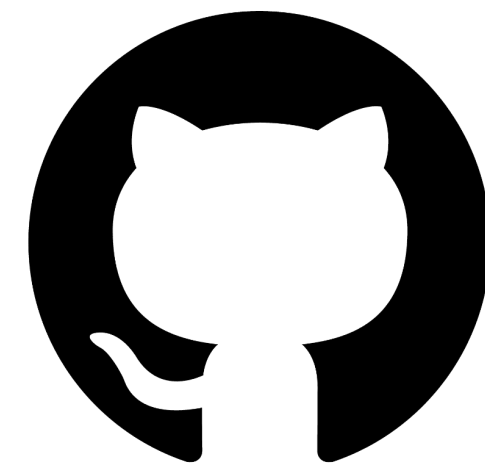
CONSUMER APPS

4TH EDITION

2025



Product Hunt



GitHub



Google Cloud



Azure

aws



Demo

Feel free to follow along!

Staying Current

Goals:

- Explore sources and tools for staying current on the latest AI research and tools

Links:

- Hugging Face [Trending Papers](#)
- MAD Landscape ([Full Resolution](#) / [Interactive](#))

Reflect + Discuss:

- Share your thoughts

How do you keep up with AI news or trends?

AI Predictions

A futuristic AI prediction visualization. In the center, a glowing crystal ball sits on a dark, conical pedestal. The crystal ball is filled with a digital map of the world, composed of numerous small, glowing blue dots. Radiating from the base of the crystal ball are several bright, blue, fiber-optic-like lines. Surrounding the crystal ball are various data visualization elements, including bar charts, line graphs, pie charts, and network diagrams, all rendered in a light blue, semi-transparent style. The background is a modern office interior with large windows, tables, and chairs, all slightly blurred to emphasize the central AI prediction theme.

10 predictions for the next 12 months

- ▶ 1. A major retailer reports >5% of online sales from agentic checkout as AI agent advertising spend hits \$5B.
- ▶ 2. A major AI lab leans back into open-sourcing frontier models to win over the current US administration.
- ▶ 3. Open-ended agents make a meaningful scientific discovery end-to-end (hypothesis, expt, iteration, paper).
- ▶ 4. A deepfake/agent-driven cyber attack triggers the first NATO/UN emergency debate on AI security.
- ▶ 5. A real-time generative video game becomes the year's most-watched title on Twitch.
- ▶ 6. "AI neutrality" emerges as a foreign policy doctrine as some nations cannot or fail to develop sovereign AI.
- ▶ 7. A movie or short film produced with significant use of AI wins major audience praise and sparks backlash.
- ▶ 8. A Chinese lab overtakes the US lab dominated frontier on a major leaderboard (e.g. LMArena/Artificial Analysis).
- ▶ 9. Datacenter NIMBYism takes the US by storm and sways certain midterm/gubernatorial elections in 2026.
- ▶ 10. Trump issues an executive order to ban state AI legislation that is found unconstitutional by SCOTUS.

What are your AI predictions for the next 12 months?